Regular Expressions (RegEx)

It is a special sequence of characters that uses a search pattern to find a string or set of strings. It can detect the presence or absence of a text by matching it with a particular pattern, and also can split a pattern into one or more sub-patterns. Python provides a re module that supports the use of regex in Python.

In [ ]:
```
Python has a built-in package called re, which can be used to work with Regular Expressions.

Import the re module:


Function          Description
findall    Returns a list containing all matches
search     Returns a Match object if there is a match anywhere in the    string
split      Returns a list where the string has been split at each match
sub        Replaces one or many matches with a string
```

In [ ]:
```
Character         Description                          Example
[]           A set of characters                       "[a-m]"
\         Signals a special sequence
        (can also be used to escape special characters)"\d"
.        Any character (except newline character)   "he..o"
^              Starts with                             "^hello"
$              Ends with                               "planet$"
*              Zero or more occurrences                "he.*o"
+              One or more occurrences                  "he.+o"
?              Zero or one occurrences                  "he.?o"
{}          Exactly the specified number of occurrences "he.{2}o"
|              Either or                               "falls|stays"
()           Capture and group
```

In [3]:
```
# Program to extract numbers from a string

import re

string = 'hello 12 hi 89.1 Howdy 344'
pattern = '\d+'

result = re.findall(pattern, string)
print(result)

# Output: ['12', '89', '34']
```

```
['12', '89', '1', '344']
```

In [8]:
```
import re

txt = "The rain in Spain"

#Find all lower case characters alphabetically between "a" and "m":

x = re.findall("[a-m]", txt)
print(x)
```

```
['h', 'e', 'a', 'i', 'i', 'a', 'i']
```

In [6]:
```
import re

txt = "That will be 59 dollars"

#Find all digit characters:

x = re.findall("\d+", txt)
print(x)
```

```
['59']
```

In [5]:
```
import re

txt = "hello planet hejko"

#Search for a sequence that starts with "he", followed excactly 2 (any) characters, and an "o":

x = re.findall("he.{2}o", txt)

print(x)
```

```
['hello', 'hejko']
```

In [6]:
```
import re
s = 'CSE portalfor Btech'

match = re.search('portal',s)
print(match)
print('Start Index:', match.start())
print('End Index:', match.end())
```

```
<re.Match object; span=(4, 10), match='portal'>
Start Index: 4
End Index: 10
```

In [10]:
```
import re

#Split the string at every white-space character:

txt = "The rain in Spain"
x = re.split("\s", txt)
print(x)
```

```
['The', 'rain', 'in', 'Spain']
```

In [9]:
```
import re

#Split the string at the first white-space character:

txt = "The rain in Spain"
x = re.split("\s", txt, 1)
print(x)
```

```
['The', 'rain', 'in', 'Spain']
```

In [14]:
```
import re

#Replace all white-space characters with the digit "9":

txt = "The rain in Spain"
x = re.sub("\s", "9", txt)
print(x)
```

```
The9rain9in9Spain
```

In [9]:
```
import re

txt = "The rain in Spain"
x = re.sub("\s", "9", txt, 2)
print(x)
```

```
The9rain9in Spain
```

In [ ]:
```
# The Match object has properties and methods used to retrieve information about the search, and the result:

.span() returns a tuple containing the start-, and end positions of the match.
.string returns the string passed into the function
.group() returns the part of the string where there was a match
```

In [11]:
```
import re

#Search for an upper case "S" character in the beginning of a word,
#and print its position:

txt = "The rain in Spain"
x = re.search(r"\bS\w+", txt)
print(x)
print(x.span())
```

```
<re.Match object; span=(12, 17), match='Spain'>
(12, 17)
```

In [ ]: